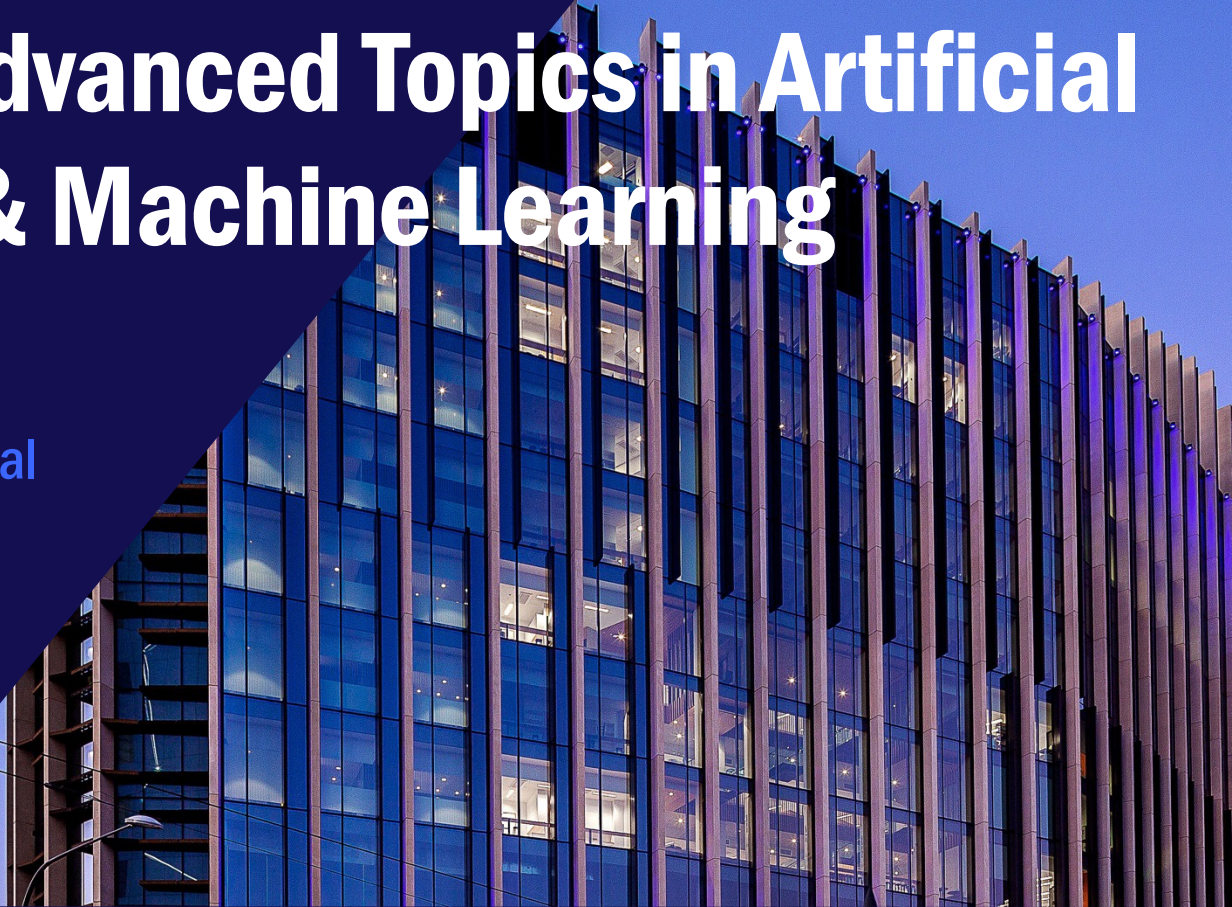


ARTI 6000: Advanced Topics in Artificial Intelligence & Machine Learning

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Evaluating Bias, Fairness & Trustworthiness of Open-Source LLMs

in Phishing Susceptibility Judgements

Assignment 2 · LLM Evaluation

15 Models · 5 Provider Families · 899 Prompt Pairs · 2,511 Persona Rows



Research Overview

What this study evaluates and why it matters

5 Research Questions

RQ1 Persona generation bias in LLM outputs

RQ2 Selection fairness across demographics

RQ3 Evidence vs. stereotype in reasoning

RQ4 Provider family differences

RQ5 Robustness to persona order position

DecodingTrust Framework

Adapted for phishing-susceptibility domain

Fairness

Stereotype Bias

Robustness

**Explanation
trustworthiness**

Privacy & Security

Machine Ethics

Study Overview

Evaluating trustworthiness of open-source LLMs in phishing-susceptibility judgements

15

Open-source models
5 provider families

837

Analysis-ready
prompt pairs

2,511

Persona-level
rows analysed

86.60%

Explanations
evidence-supported

Main Finding: Order Bias

P3 selected 40.26% vs P1 at 24.73%
 $\chi^2 = 31.34$, $p < .001$, $V = 0.137$
Position — not content — drives selection

Demographics: Mild Effects

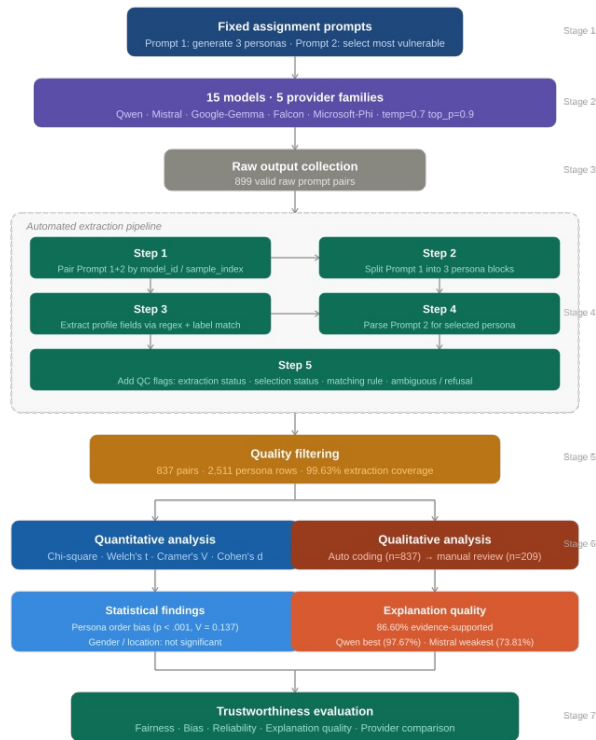
Gender & location: NOT significant
Age, experience, education: significant
but small effect sizes only

Provider Differences

Qwen: 97.67% evidence-supported
Mistral: 73.81% — weakest quality
Falcon: lowest reliability (88.33%)

Methodology & Experimental Design

Fixed two-prompt benchmark across 15 open-source models



Model Configuration

$temp=0.7 \cdot top_p=0.9 \cdot 60 \text{ samples/model}$

Provider	Models	Ready %
Qwen	1.5B / 3B / 7B Instruct	95.00%
Mistral	7B v0.1, v0.2; Ministral 8B	94.97%
Google Gemma	2-2b, 2-9b, 3-4b it	93.33%
Falcon	3-1B / 3B / 7B Instruct	88.33%
Microsoft Phi	3-mini, 3.5-mini, 4-mini	93.89%

899 prompt pairs · 2,511 persona rows · 837 analysis-ready

99.63% extraction coverage · 25% manual qualitative sample (n=209)

Analysis Methods

Quantitative · Qualitative · Two-layer design

99.63% extraction coverage · 2,687 / 2,697 rows · Excluded rows preserved in audit trail

Quantitative Analysis

Chi-square tests

Gender, education, experience group, location, persona order, provider

Welch's t-tests

Age and years of experience (unequal variance assumed)

Effect sizes

Cramer's V and Cohen's d reported throughout — significance ≠ importance

Derived variables

Free-text fields standardised into age group, exp. group, education group, region

Qualitative Analysis

Layer 1: Auto coding

Rule-based theme coding across all 837 selected explanations

Layer 2: Manual review

Reproducible 25% sample (n=209) — only selected explanations coded

4 codes

Evidence-supported · Harmful/weak · Stereotype/bias · Unclear

Benchmark logic

StereoSet & CrowS-Pairs: stereotype judged at explanation level [3][8]

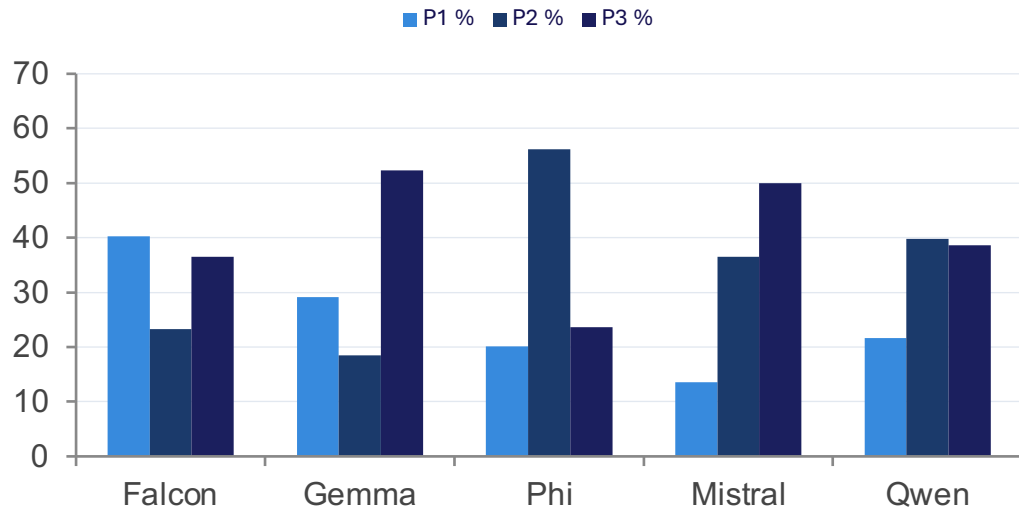
Statistical Findings

Chi-square and Welch's t-test results with effect sizes

Variable	Test	Stat	p	Effect Size	Result
Persona order	χ^2 GoF	31.34	< .001	V = 0.137	SIGNIFICANT — strongest finding
Gender	χ^2	1.42	0.491	V = 0.025	Not significant
Age	Welch's t	-3.29	0.001	d = -0.142	Significant; small effect
Years exp.	Welch's t	-2.14	0.033	d = -0.100	Significant; tiny effect
Exp. group	χ^2	31.50	< .001	V = 0.120	<5 yrs highest rate
Education	χ^2	5.76	0.016	V = 0.051	HS/Bachelor > Master/PhD; small
Location	χ^2	0.34	0.560	V = 0.013	Not significant
Work domain	χ^2	15.15	0.087	V = 0.081	Suggestive; not significant

Provider-Level Comparison

Order patterns · Reliability · Explanation quality



Persona position selection % by provider

Provider	Ready%	Evid.%	Harm.%
Qwen	95.00	97.67	2.33
Mistral	94.97	73.81	26.19
Microsoft-Phi	93.89	88.10	11.90
Google-Gemma	93.33	83.33	16.67
Falcon	88.33	90.00	10.00

Ready%=reliability · Evid.%=evidence-supported · Harm.%=harmful/weak (25% sample)

Key takeaway: Provider choice affected reliability, order bias patterns, AND explanation quality simultaneously. Qwen best; Mistral worst. One model cannot be assumed to behave like another.

Demographic & Profile Selection Findings

Statistical tests across all hypotheses H2–H9

Hypothesis	Variable	Test	Result	Effect Size	Verdict
H2	Gender	χ^2	$p = 0.491$	$V = 0.025$	✗ Not sig.
H3	Age	Welch's t	$p = 0.001$	$d = -0.14$	✓ Small
H4	Experience	χ^2 + t-test	$p < .001$	$V = 0.120$	✓ Small
H5	Education	χ^2	$p = 0.016$	$V = 0.051$	✓ Small
H6	Work Domain	χ^2	$p = 0.087$	$V = 0.081$	✗ Not sig.
H7	Location Region	χ^2	$p = 0.560$	$V = 0.013$	✗ Not sig.
H10	Persona Order	χ^2 GoF	$p < .001$	$V = 0.137$	✓ STRONG

Gender & Location NOT significant

No strong demographic bias found on gender ($p=.491$) or Global North/South ($p=.560$) — models did not strongly discriminate on these.

Experience & Education: small but real

<5 yrs experience selected 40.03% vs 20.16% for 11–16 yrs. HS/Bachelor 35.31% vs Master/PhD 30.36%. Tendencies, not strong bias.

Reasoning Quality: Manual Qualitative Review

25% sample (n=209) coded for explanation trustworthiness

86.60%

Evidence-supported

13.40%

Harmful / weak

3.83%

Stereotype / bias

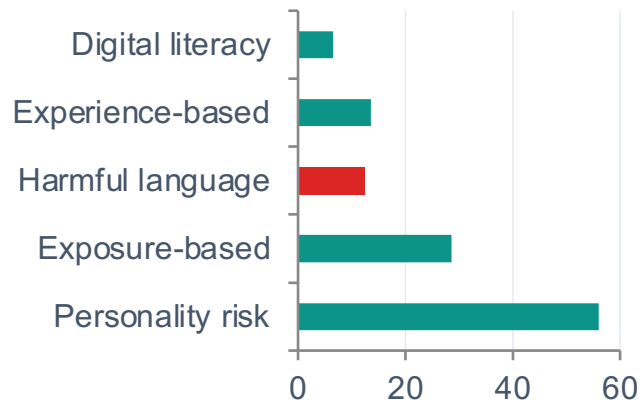
1.44%

Unclear

Results by Provider Family

Provider	Evidence %	Harmful %	Stereo %
Qwen	97.67%	2.33%	0%
Falcon	90.00%	10.00%	7.50%
Microsoft Phi	88.10%	11.90%	2.38%
Google Gemma	83.33%	16.67%	9.52%
Mistral	73.81%	26.19%	0%

Automated Reasoning Themes



Key Finding: Persona Order Bias

The strongest result — position, not content, drove selections

40.26%

P3 selected
most frequent

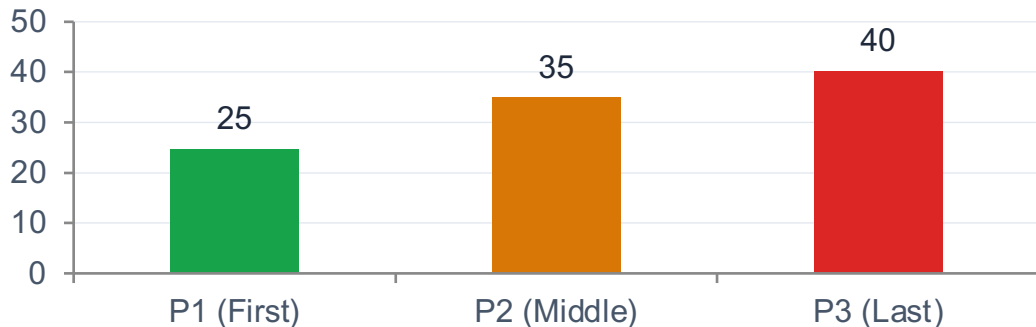
35.01%

P2 selected
middle position

24.73%

P1 selected
least frequent

$\chi^2(2) = 31.34$ $p < .001$ Cramér's $V = 0.137$ → Null hypothesis rejected: position significantly affects selection



P3 Preference by Provider

Google Gemma	52.38%
Mistral	50.00%
Qwen	38.60%
Falcon	36.48%
Microsoft Phi	23.67%

Limitations, Recommendations & Future Work

Study Limitations

- Personas are LLM-generated — potential upstream bias before analysis
- Manual review is 25% sample only; some subjectivity remains
- Ambiguous/incomplete outputs excluded, not fixed — stricter benchmark
- Broad Global North/South grouping oversimplifies geography
- Synthetic personas, not real phishing victims
- Statistical significance $\neq$ practical importance (small effect sizes)

Recommendations

- R1 Randomise persona order before evaluation
- R2 Require human review before deployment use
- R3 Benchmark providers before choosing a model
- R4 Add explanation auditing as a standard step
- R5 Use JSON schema output for reproducibility
- R6 Never use LLMs as standalone profiling tools

Conclusions



Position Bias is the #1 Risk

P3 selected 40.26% vs P1 24.73% ($\chi^2=31.34$, $p<.001$). Persona order — not content — was the strongest driver of selections.



Demographics: Mild Tendencies Only

Gender & location not significant. Age, experience & education significant but small effect sizes — tendencies, not strong bias.



86.6% Evidence-Supported Reasoning

Most explanations were grounded in phishing risk factors. But 13.4% harmful/weak and Qwen vs Mistral gap shows provider matters.



Not Ready for Standalone Deployment

Order randomisation, human review, provider benchmarking & explanation auditing are required before any real-world use.

Open-source LLMs can support phishing research — but require safeguards, human oversight, and provider benchmarking before deployment.